MIL-F-23138(Wep)
3 January 1962
Superseding
OS 7187B
10 March 1960

MILITARY SPECIFICATION

FLARE, TRACKING, MK 23 MOD 0

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of the Flare, Tracking, Mk 23 Mod 0 and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specifications for

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1370

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MIL-STD-350	Jolt Test for use in Production of Fuzes
MIL-STD-351	Jumble Test for use in Production of Fuzes
MIL-STD-352	Forty (40) Foot Drop Test for use in Production of Fuzes
MIL-STD-354	Temperature and Humidity Test for use in Production of Fuzes

DRAWINGS

Bureau of Ordnance

LD 479375	Flare, Tracking, Mk 23 Mod 0
LD 479399	Container (Inner) for Tracking Flare Mk 23 Mod 0
LD 479403	Container (Outer) for Tracking Flare Mk 23 Mod 0

And all drawings, standards and specifications listed on the above LD's and drawings.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this document. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

JOURNAL OF THE OPTICAL SOCIETY OF AMERICA

D.B. Judd	The 1931 I.C.I. Standard Observer and Coordinate System for Colorimetry (1933)(Vol. 23, p. 359)

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George C. Szikiai — A Tristimulus Photometer, Vol. 41, No. 5, p. 321, (1951)

(Application for copies of the above should be addressed to American Institute of Physics, 335 East 45 St., New York 17, New York.)

AMERICAN SOCIETY FOR TESTING MATERIALS

Symposium on Color — Its specification and use in Evaluating the Appearance of Materials (1941)

(Application for copies of the above should be addressed to American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

U. S. DEPARTMENT OF COMMERCE

NATIONAL BUREAU OF STANDARDS

Circular C-429 — Photoelectric Tristimulus Colorimetry with Three Filters, R. S. Hunter (1942)

(Applications for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 — Interstate Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite the latest edition and supplements)

3. REQUIREMENTS

3.1 Description. The Flare, Tracking, Mk 23 Mod 0 is a missile tracking flare electrically initiated by the missile in flight.

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3.2 General requirements.

3.2.1 The Flare, Tracking shall be manufactured, tested, inspected and packaged in accordance with the contract documents, applicable drawings and this specification.

3.2.2 Definition. The term "flare" as used in this specification, unless otherwise specified, shall mean Flare, Tracking, Mk 23 Mod 0 (BuOrd LD 479375).

3.2.3 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of flares shall be furnished by the contractor at his expense. The sample shall be manufactured, inspected and evaluated as specified in 4.3.

3.2.4 Conflicting requirements. Any conflict arising between this specification and any specification, publication or drawing cited herein shall be referred in writing to the procuring agency for clarification.

3.2.5 Material. Unless otherwise specified in the contract or order, all materials and purchased parts entering into the final assembly of the flare shall conform strictly to the detailed requirements indicated on the drawings and specifications listed on LD 479375.

3.3 Performance and product characteristics requirements.

3.3.1 Electrical requirements.

3.3.1.1 Insulation resistance. The insulation resistance of the flare between the shorted contact terminals and the flare shell assembly (1520424) when tested as specified in 4.5.2.1, shall be equal to or greater than 10 megohms under an applied test potential of 100 $\pm$ 10 volts dc.

3.3.1.2 Circuit resistance. The resistance of the electrical circuit measured between the contact terminals shall be 0.35 ohms minimum to 0.65 ohms maximum when tested as specified in 4.5.3.1.

3.3.1.3 Non-firing current. The flare when tested as specified in 4.5.4.1, shall not function when a current of .25 $\pm$ .05 amperes dc flows through the firing circuit for a minimum time of 5 seconds.

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3.3.2 Functioning requirements.

3.3.2.1 Static functioning. The flare upon receiving a current of 3.00 ± .30 amperes dc when tested as specified in 4.5.5.1 shall:

a. Develop a minimum of 15,000 candlepower within 1.5 seconds after initiation.

b. Develop a minimum of 70,000 candlepower within 3 seconds after initiation.

c. Develop a minimum average of 70,000 candlepower for the 20 seconds following initiation.

d. Between 3 seconds and 20 seconds after initiation the candlepower shall not fall below 50,000 for more than 0.2 seconds at any one time.

e. Produce a red flame with a dominant wave length of 605 to 625 millimicrons and a purity of 55 percent minimum.

3.3.2.2 Acceleration and combustion. The flare shall burn for a minimum of 20 seconds and shall not produce or break up into particles, any one of which weighs more than 0.1 grams (excluding the closing disc), when subjected to the acceleration and combustion test of 4.5.6.1.

3.4 Durability and environmental requirements.

3.4.1 Safety. The flare shall not come apart, burn, explode or become unsafe to handle and dispose of when subjected to the jolt, jumble and drop tests as prescribed in 4.6.5, 4.6.6 and 4.6.7 respectively.

3.4.2 Vibration. The flare shall withstand the vibration test specified in 4.6.8 without showing evidence of decomposition, burning or mechanical failure. After the vibration test the flare shall meet the functioning requirements of 3.3.2.1 when tested as specified in 4.5.5.1.

3.4.3 Shock. When subjected to the shock test of 4.6.10 the flare shall show no signs of damage, and while burning shall not produce or break up into particles any one of which weighs more than 0.1 grams (excluding the closing disc).

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3.4.4 Temperature, pressure and humidity. The flare shall meet the functioning requirements of 3.3.2.1 after being subjected to the temperature, pressure, humidity cycling tests prescribed in 4.6.11.

3.4.5 Low pressure. The flare, when tested as specified in 4.6.12, shall ignite within 1.5 seconds and burn for a minimum of 20 seconds from initiation while being subjected to a pressure of 11.0 ± 0.5 inches of mercury absolute (corresponding approximately to an altitude of 25,000 ft.).

3.4.6 Salt spray resistance. The flare shall withstand the salt spray test specified in 4.6.13 without showing visible evidence of corrosion.

3.4.7 Package sealing requirements.

3.4.7.1 Leakage. The sealed can containing a flare, when tested as specified in 4.6.14.1, shall withstand a vacuum of 3.0 psig minimum without evidence of leakage.

3.4.7.2 Temperature and humidity. The flare sealed in the specified container shall withstand the temperature humidity cycling tests specified in 4.6.14.2 without loss of functional characteristics, meeting the requirements of 3.3.2.1 when tested as specified in 4.5.5.1.

3.5 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. There shall be no imperfections or deviations from the specifications affecting the functioning or assembly of the product. Where applicable, all materials shall be free of pits, corrosion, cracks, spalls, rough edges, etc. All components shall be free of chips, dirt, grit or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. Unless otherwise required by the procuring activity, the contractor is responsible for the performance of all tests and examinations to assure that the flare delivered to the Government is in full accordance with applicable requirements. In addition, the contractor is required to maintain records of tests and examinations performed and to make these records available to the Government representative upon

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4.2 Government inspection. The government representative will examine the contractor's records and processes and will witness or perform examinations and tests as are deemed necessary to assure that components and assemblies of the flare are in accordance with all applicable requirements.

4.2.1 Sampling. Unless otherwise specified and when applicable the sampling plans and procedures used by the Government inspector in the determination of acceptability of products submitted by the contractor for Government inspection, shall be in accordance with the provisions of Standard MIL-STD-105.

4.2.2 Lot. As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in inspection lots shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purposes.

4.3 Preproduction sample. As specified in 3.2.3, a sample of 75 flares shall be manufactured using the same procedures, processes and materials and at the same location proposed for the production lot. Upon inspection and notation of any deficiencies by the inspector, the samples shall be forwarded, together with a report of inspection to an activity designated by the procuring agency for final inspection and acceptance as specified in 4.6 (see figure 1). Samples shall be prepared for delivery as specified in section 5.

4.3.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the Government when production under a new contract by the same contractor, at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3.2 Periodic production sample. Unless otherwise specified by the procuring agency, a sample of 75 flares shall be drawn from one lot selected at random from each five consecutive production lots by the Government inspector for submission to an activity designated by the procuring agency for inspection and acceptance as specified in 4.6 (see figure 1). Samples shall be prepared for delivery as specified in section 5.

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4.4 Production lot. Unless otherwise specified by the procuring agency, each production lot shall consist of 1200 flares plus those required for test purposes. All units required for test purposes shall be at the expense of the contractor.

4.5 Acceptance tests.

4.5.1 Test equipment. Unless otherwise specified in the procurement documents, all test equipment shall be furnished by the contractor. The contractor shall certify to the Government inspector that test equipment, including instrumentation, has been calibrated before proceeding with production. Test equipment that applies any electrical charge to the ignition circuit of the flare shall provide for the safety precautions of 6.3.1.

4.5.2 Insulation resistance test. Samples shall be selected from each lot in accordance with Standard MIL-STD-105, Inspection Level 1, AQL 0.25 percent defective.

4.5.2.1 Test procedures shall be in accordance with Standard MIL-STD-202, Method 302. The test temperature and relative humidity shall not exceed 79 degrees F and 80 percent respectively. (To facilitate production ambient conditions may exceed the prescribed limits for acceptance, but rejection of flares shall be based on tests conducted as prescribed.) Conformance to the requirements of 3.3.1.1 shall determine acceptability.

4.5.3 Circuit resistance test. Samples shall be selected from each lot in accordance with Standard MIL-STD-105, Inspection Level 1, AQL 0.25 percent defective.

4.5.3.1 Test procedures shall be in accordance with MIL-STD-202, Method 303. The test current shall be less than 20 milli-amperes dc. Conformance to the requirements of 3.3.1.2 shall determine acceptability.

4.5.4 Non-firing current test. Samples shall be selected from each lot in accordance with MIL-STD-105, Inspection Level L8, AQL 0.40 percent defective. Flares used in this test shall not be returned to the lot but may be used to fill the sample requirements of 4.5.6.

4.5.4.1 Test procedures. The applied current and duration of the test shall be as specified in 3.3.1.3. A flare which functions during the test shall be considered a defective.

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4.5.5 Static functioning test. Samples shall be selected from each lot in accordance with MIL-STD-105, Inspection Level L7, AQL 2.5 percent defective.

4.5.5.1 Test procedure. The flare shall be mounted in the fire place vertically with the closing disc up and 4.0 ± .5 inches below the photometric axis. There shall be a minimum opening of 30 inches into the fire place above the photometric axis. The tunnel shall have non-reflecting surfaces and four equally spaced baffles. Light intensity and color shall be measured by the Tristimulus Photometer as described in the "Journal of the Optical Society of America" Volume 41 (1951) pp 321-323, (or equal). Calibration of the photometer shall be made using a projector lamp, calibrated for luminous intensity and color by the National Bureau of Standards, Washington, D. C. Using the Tristimulus Photometer, the luminous intensity, the dominant wave length and the percent of purity of the red flame produced by the burning flare shall be measured and calculated in accordance with I.C.I. (International Commission on Illumination) Tristimulus Color System, Journal of the Optical Society of American, Volume 23 (1933) p 359; ASTM Symposium on Color (1941); and National Bureau of Standards Circular C-429.

The temperature of the flare at time of ignition shall be 80 ± 10 degrees F. A direct current of 3.00 ± .30 amperes from a remote observation station shall be applied to the ignition circuit of the flare. The flare shall function and meet the requirements of 3.3.2.1 (a) through (e).

4.5.6 Acceleration and combustion test. Samples shall be selected from each lot in accordance with MIL-STD-105, Inspection Level L6, AQL 1.0 percent defective. Flares from non-firing test of 4.5.4 may be used to fill the sample requirements. Flares shall meet the requirements of 3.3.2.2 when subjected to the test of 4.5.6.1.

4.5.6.1 Test procedure. The flares shall be tested in a centrifuge surrounded by a mesh enclosure so that particles ejected by the burning flare are retained for examination. The flare shall be mounted in a position which produces an acceleration perpendicular to the longitudinal axis of the flare. The flare shall be attached to the centrifuge by a band support approximately on the center of the length of the flare body. The centrifuge shall be brought up to a rotational velocity which produces an acceleration of 35 gravity units and this shall be maintained for a minimum of 15 seconds. The flare shall then be

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rotated 90 degrees so the acceleration is applied parallel to the axis with the closing disc of the flare outward and in such a manner that the force of acceleration is applied to the quick disconnect fitting. The centrifuge shall be brought up to a rotational velocity which produces an acceleration of 100 gravity units. The flare shall then be ignited and the acceleration maintained for no less than 5 seconds and by 10 seconds the acceleration shall be reduced to develop not more than 35 gravity units. The burning time of the flare shall be recorded and the ejected particles removed for examination.

4.6 Preproduction and periodic sample tests. The tests shall be conducted on the basis of the preproduction sample and the periodic lot samples as specified in 4.3.

4.6.1 Insulation resistance test. Ten flares from the sample shall be subjected to the insulation resistance test as specified in 4.5.2.1. Failure of one or more of the flares to meet the requirements of 3.3.1.1 shall reject the sample. The flares used for this test shall be used for the test of 4.6.2.

4.6.2 Circuit resistance test. Ten flares from the test of 4.6.1 shall be tested for the resistance of the ignition circuit as specified in 4.5.3.1. Failure of one or more of the flares to meet the requirements of 3.3.1.2 shall reject the sample. The flares used for this test shall be used for the static function test of 4.6.4.

4.6.3 Non-firing current test. Ten flares from the sample shall be subjected to the non-firing current test of 4.5.4.1. Failure of one or more flares to meet the requirements of 3.3.1.3 shall reject the sample. The flares that do not function in this test shall be used for the tests of 4.6.9 and 4.6.10.

4.6.4 Static functioning test. The ten flares from the test of 4.6.2 shall be subjected to the functioning test of 4.5.5.1. Failure of two or more of the flares to meet the requirements of 3.3.2.1, (a) through (e) shall reject the sample. The data obtained from this test shall then be grouped in 4.6.15 to determine overall functional acceptability.

4.6.5 Jolt test. Five flares from the sample shall be subjected to the jolt test as prescribed in MIL-STD-350. Each flare shall be rigidly assembled to the arm of the jolt machine. Failure of one or more flares to meet the requirements of 3.4.1 shall reject the sample.

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4.6.6 Jumble test. Five flares from the sample shall be subjected to the jumble test prescribed in MIL-STD-351. Failure of one or more of the flares to meet the requirements of 3.4.1 shall cause rejection of the sample.

4.6.7 Forty-foot drop test. Five flares from the sample shall be subjected to the forty-foot drop test prescribed in MIL-STD-352. Failure of one or more of the flares to meet the requirements of 3.4.1 shall cause rejection of the sample.

4.6.8 Vibration test. Five flares from the sample shall be subjected to the vibration test prescribed in 4.6.8.1. Failure of one or more of the flares to meet the durability requirements of 3.4.2 or failure of two or more of the flares to meet the functioning requirements of 3.3.2.1 as specified in 3.4.2 shall reject the sample. Data obtained from this test shall then be grouped in 4.6.15 to determine overall functional acceptability.

4.6.8.1 Test procedure. The vibration test equipment shall be as specified in MIL-E-5272 or similar equipment approved by the procuring agency. Each flare shall be subjected to the following vibration schedule.

a. The flares shall be mounted directly on the vibration table with the direction of vibration parallel to the longitudinal axis of the flare. The flare shall be subjected to one hour of constantly applied vibration frequency cycling between 10 and 60 cycles per second in 15 minute cycles at an applied double amplitude of 0.060 inches or an applied acceleration of 9 gravity units whichever is the limiting value.

b. Repeat step a. with the direction of vibration applied laterally to the longitudinal axis of the flare.

c. The flare shall then be subjected for one hour to a constantly applied vibratory acceleration of 5 gravity units along the longitudinal axis with the frequency of cycling between 60 and 500 cycles per second in 15 minute cycles.

d. Repeat step c. with the direction of vibration applied laterally to the longitudinal axis of the flare.

e. The flares shall then be subjected for one hour to a constantly applied vibratory acceleration parallel to the longitudinal axis of the flare of 3 gravity units

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with a frequency cycling between 500 and 1200 cycles per second in 15 minute cycles.

f. Repeat step e. with the direction of vibration applied laterally to the longitudinal axis of the flare.

After the vibration schedule, the flares shall be examined visually for conformance to the durability requirements of 3.4.2 then tested functionally as specified in 4.5.5.1 for conformance to the functioning requirements of 3.3.2.1.

4.6.9 Acceleration and combustion test. Five flares from the sample shall be subjected to the acceleration test prescribed in 4.5.6.1. Failure of one or more of the flares to meet the burning time and combustion requirements specified in 3.3.2.2 shall reject the sample.

4.6.10 Shock test. Five flares from the sample shall be subjected to the simulated buster shock test. The flare shall be rigidly mounted on the shock testing machine with the terminal end facing the direction of motion. The flare shall be subjected to a shock of 100 $\pm$ 5 gravity units parallel to the longitudinal axis, the shock having a time duration of 11 milliseconds with the maximum gravity unit value occurring at approximately 3 milliseconds. After the shock the flare shall show no signs of damage and shall be mounted in a test chamber where any particles ejected while burning can be recovered and examined. The flares shall be ignited by the application of a dc current of 2.7 amperes or greater. Failure of one or more of the flares to meet the requirements of 3.4.3 shall reject the sample.

4.6.11 Temperature, humidity, pressure test. Fifteen flares from the sample shall be subjected to the temperature, humidity, pressure tests as specified in 3.4.4. The incidence of ignition or decomposition which makes a flare unsafe to use in any phase of the test shall cause rejection of the sample.

4.6.11.1 High to low temperature test. Ten flares from the sample of 4.6.11 shall be stabilized at ambient temperature, then subjected to the following temperature cycles:

a. The flares shall be placed in a chamber with ambient pressure and the temperature raised to + 185 $\pm$ 5 degrees F. This temperature shall be maintained for a minimum of 4.0 hours.

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b. The flares shall be removed from the high temperature chamber and within five minutes placed in a chamber in which the temperature is -85 ± 5 degrees F. The chamber shall be evacuated to a pressure of 3.4 ± .5 inches of mercury absolute. The pressure and temperature shall be maintained for a minimum of 4.0 hours.

c. The flares shall be removed from the low temperature chamber and within five minutes placed in a chamber with an internal temperature of +185 ± 5 degrees F, and a pressure of 29.9 ± .5 inches of mercury absolute. This temperature and pressure shall be maintained for a minimum of 4.0 hours.

d. Repeat step b.

e. Repeat step c.

f. Repeat step b.

After completion of step (f) five of the flares shall be allowed to return to ambient conditions of temperature and humidity and retained for the humidity tests of 4.6.11.3. The remaining five flares shall be subjected to the functioning test of 4.5.5.1. Test procedures shall be regulated to maintain the specimen temperature at not greater than -70 degrees F at time of ignition. The incidence of two or more defectives shall cause rejection of the sample. Data obtained from this test shall then be grouped in 4.6.15 to determine overall functional acceptability.

4.6.11.2 Low to high temperature test. Subject five flares from 4.6.11 to three complete temperature cycles as described in 4.6.11.1, except that the cycling shall begin with the low temperature (b) and end with the high temperature phase (c). The flares shall be removed from the chamber and subjected to the functioning test of 4.5.5.1. Test procedures shall be regulated to maintain the specimen temperature at not less than +180 degrees F at time of ignition. The incidence of two or more functioning defectives shall cause rejection of the sample. Data obtained from this test shall then be grouped in 4.6.15 to determine overall functional acceptability.

4.6.11.3 Humidity test. The five flares which were prepared by the temperature cycle test of 4.6.11.1 shall be subjected to the humidity test. The flares shall be placed in a chamber for a period of 24 hours in which the temperature and relative humidity shall be maintained at 100 ± 5 degrees F and 94 ± 5 percent

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respectively. At the conclusion of this period the flares shall be removed from the chamber and within five minutes subjected to the functioning test of 4.5.5.1. The incidence of two or more defectives shall cause rejection of the sample. Data obtained from this test shall then be grouped in 4.6.15 to determine overall functional acceptability.

4.6.12 Low pressure functioning test. Five flares from the sample shall be tested for ignition and burning in a pressure chamber with an internal pressure of 11.0 $\pm$ 0.5 inches of mercury absolute. The flare shall be initiated by the application of a dc current of 2.7 amperes minimum. Failure of two or more flares to meet the requirements of 3.4.5 shall reject the sample.

4.6.13 Salt spray test. Five flares from the sample shall be subjected to the salt spray as specified in Federal Test Method Standard No. 151, Method 811, duration of test 24 hours as required by 3.4.6. The sample shall be rejected if one or more of the flares show evidence of corrosion as a result of the test.

4.6.14 Packaging test.

4.6.14.1 Leakage. Five flares, packaged as required by 5.1.1 shall be tested for package leakage. The sealed containers shall be submerged 3 to 4 inches below the surface of the water in a container equipped for observation under vacuum. A vacuum of 3.0 +.2 -.0 psig shall be established over the surface of the water and maintained for a minimum of 15 seconds. Leakage appears as streams of bubbles issuing from the container. Flares from the containers after the tests may be used for examination or to complete samples for other tests.

4.6.14.2 Temperature and humidity test. Five flares from the sample packaged as provided in 5.1.1 shall be subjected to the temperature and humidity test (28 days) as prescribed in MIL-STD-354. Immediately after completion of the temperature and humidity test the flares shall be removed from the containers and static function tested as prescribed in 4.5.5.1. Failure of two or more of the flares to meet the requirements specified in 3.4.7.2 shall reject the sample. Data from this test shall be grouped in 4.6.15 to determine overall functional acceptability.

4.6.15 Overall environmental functioning acceptability. The data obtained from the tests listed below shall be grouped showing the number of failures in each test to meet the functioning requirements of 3.3.2.1 (a) through (e).

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a. 4.6.8 Vibration test	5 flares
b. 4.6.11.1 High-low temperature test	5 flares
c. 4.6.11.2 Low-high temperature test	5 flares
d. 4.6.11.3 Humidity test	5 flares
e. 4.6.14.2 (Packaged) temperature humidity test	5 flares
f. 4.6.4 Static function test	10 flares

In addition to the acceptance requirements of the individual tests, failure of three or more of the 35 flares to meet the functioning requirements of 3.3.2.1 shall reject the sample.

4.7 Examination

4.7.1 Packaging, packing and marking inspection. The Government inspector shall ascertain that the packaging, packing and marking of the flares and containers conform to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.1.1.1 Unit packaging. The flares shall be packaged in accordance with BuOrd LD 479399.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Not applicable.

5.2 Packing

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.2.1.1 Twenty-four packaged flares shall be packed in accordance with BuOrd LD 479403.

5.2.2 Level B. Not applicable.

5.2.3 Level C. Not applicable.

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5.3 Marking

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.3.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.3.3 Marking of preproduction sample containers. The preproduction sample containers shall be marked as specified in 5.3 and shall also be marked with the following:

"Samples of preproduction lot. For acceptance tests in accordance with MIL (NOrd). Authorized by _____ (Insert shipping authority letter, number and date)."

6. NOTES

6.1 Intended use. The Flare, Tracking Mk 23 Mod 0 is used for tracking missiles in flight.

6.2 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.3 General safety precautions. The loading, assembly and handling of the item covered by this specification, and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precaution. Use of this specification will not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a contractor or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Bureau of Ordnance, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions pertinent to the operational effectiveness or safety during use or loading of the specified item should be included in the detail technical requirements of the specification.

6.3.1 When any electrical charge is applied to the firing circuit of the flare, the flare shall be securely mounted and shielded to prevent personal injury or property damage if the flare should ignite.

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6.3.2 All loading operations should be conducted in a neat and orderly manner.

6.3.3 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, coating, crimping, etc.

6.3.4 Personnel handling detonators, primers, delay elements, lead-ins, boosters and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

6.3.5 The exposure of explosive materials and related parts shall be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.6 All explosives and completely or partly loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards, and while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.3.7 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

6.3.7.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.3.7.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations

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6.3.7.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.3.7.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures; and where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.3.7.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices should be employed during

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manufacturing operations involving electric primers, detonators or squibs.

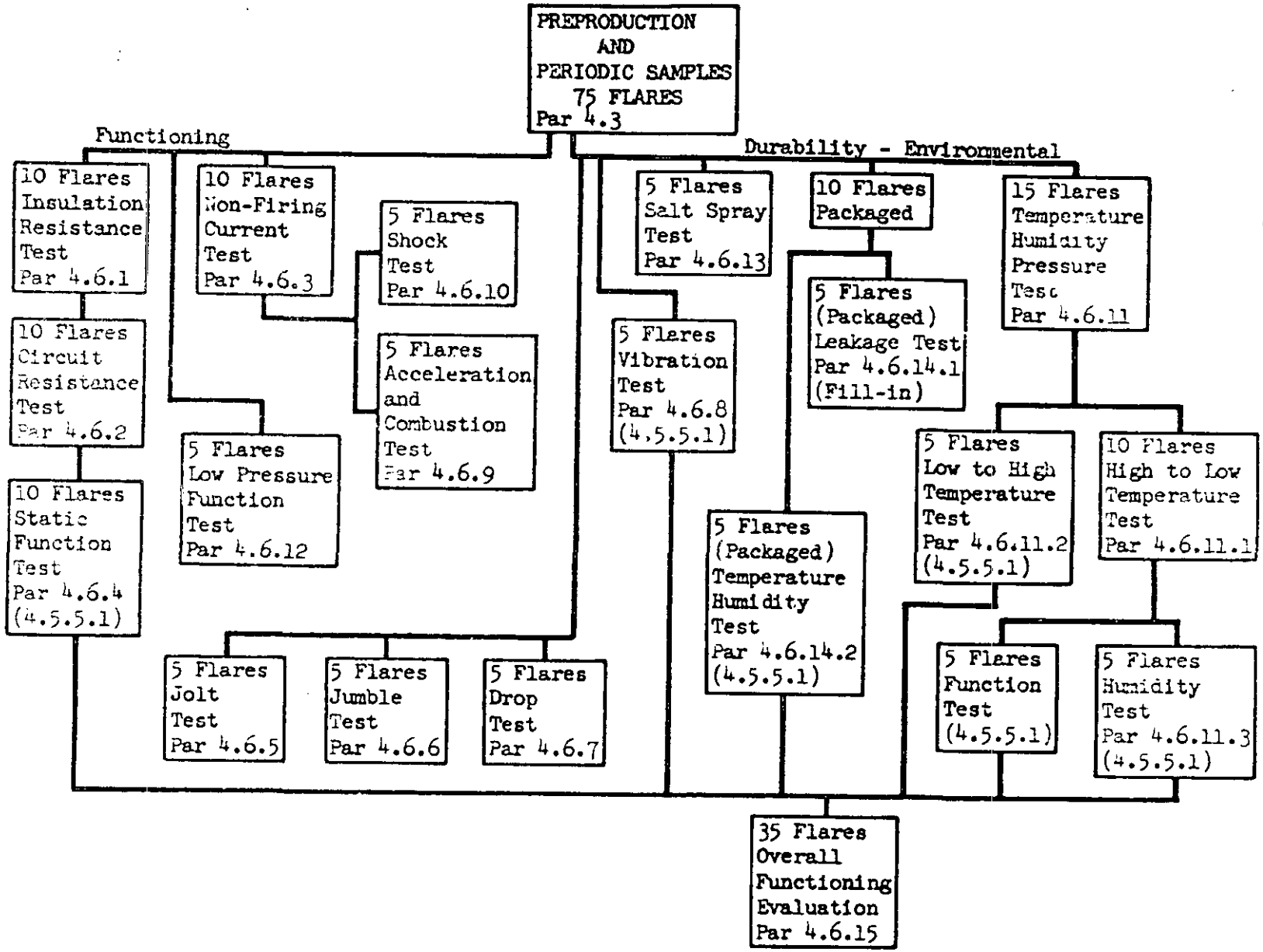
6.3.7.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.3.7.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.3.7.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

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FLOW CHART

Figure 1

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